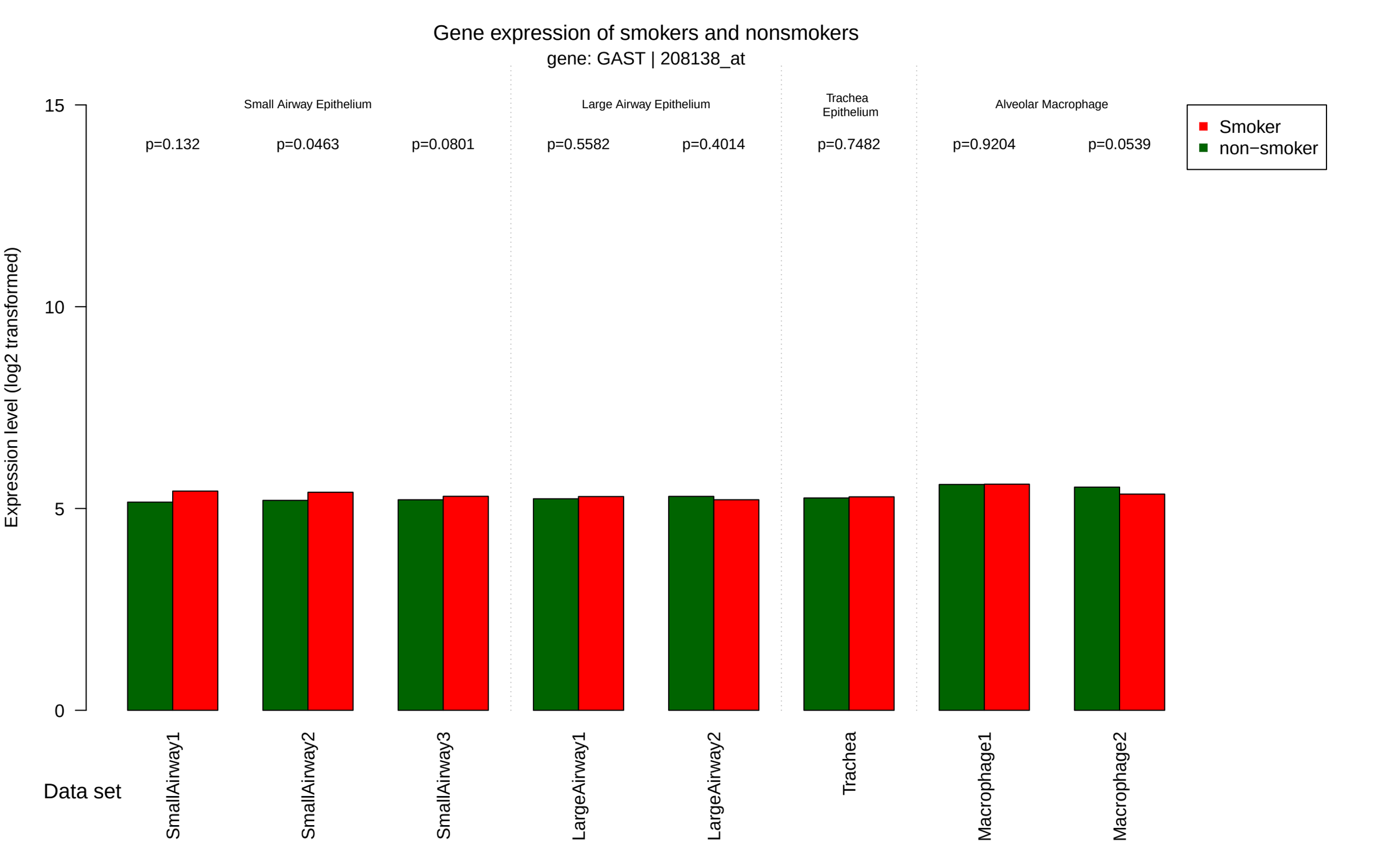
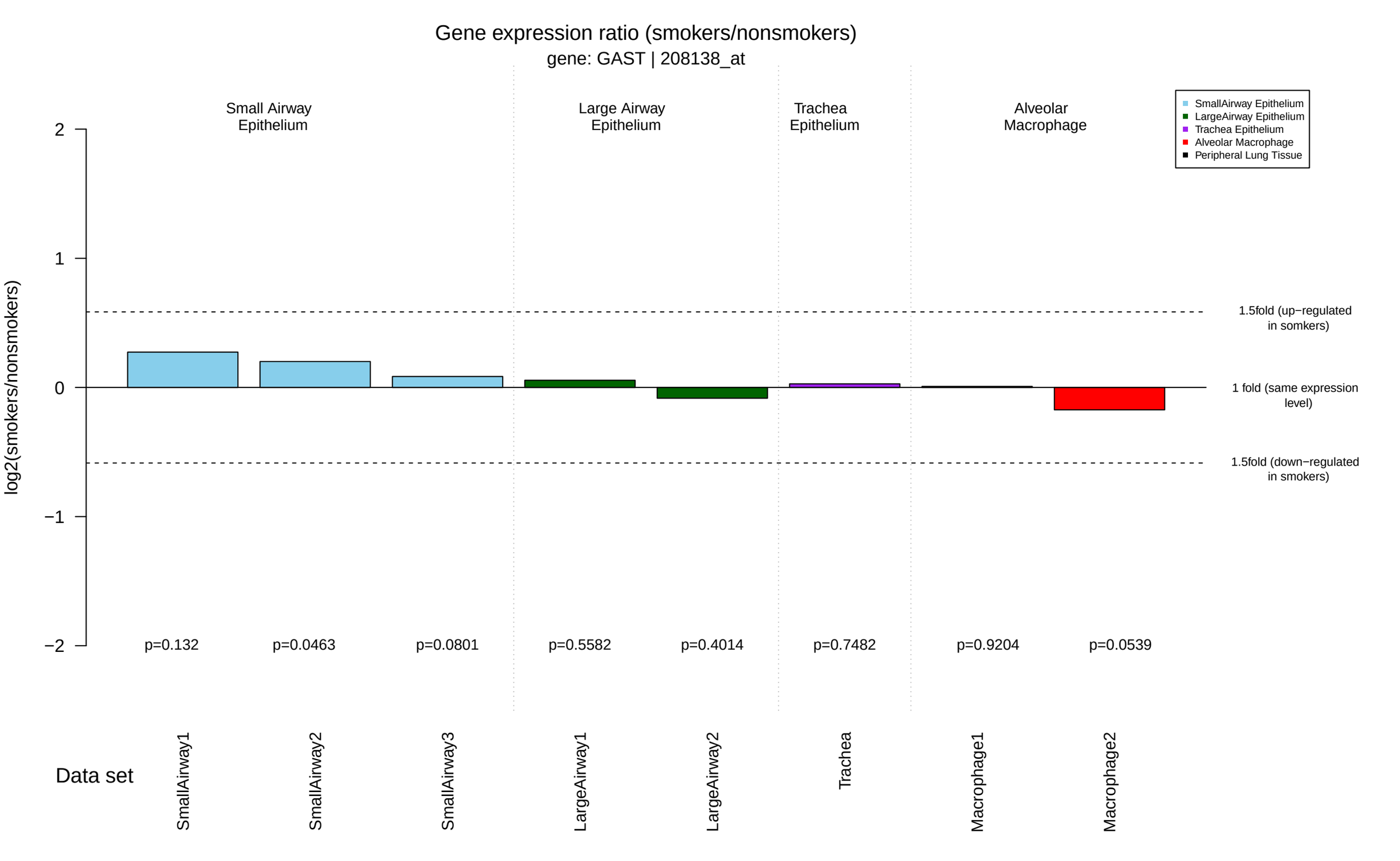
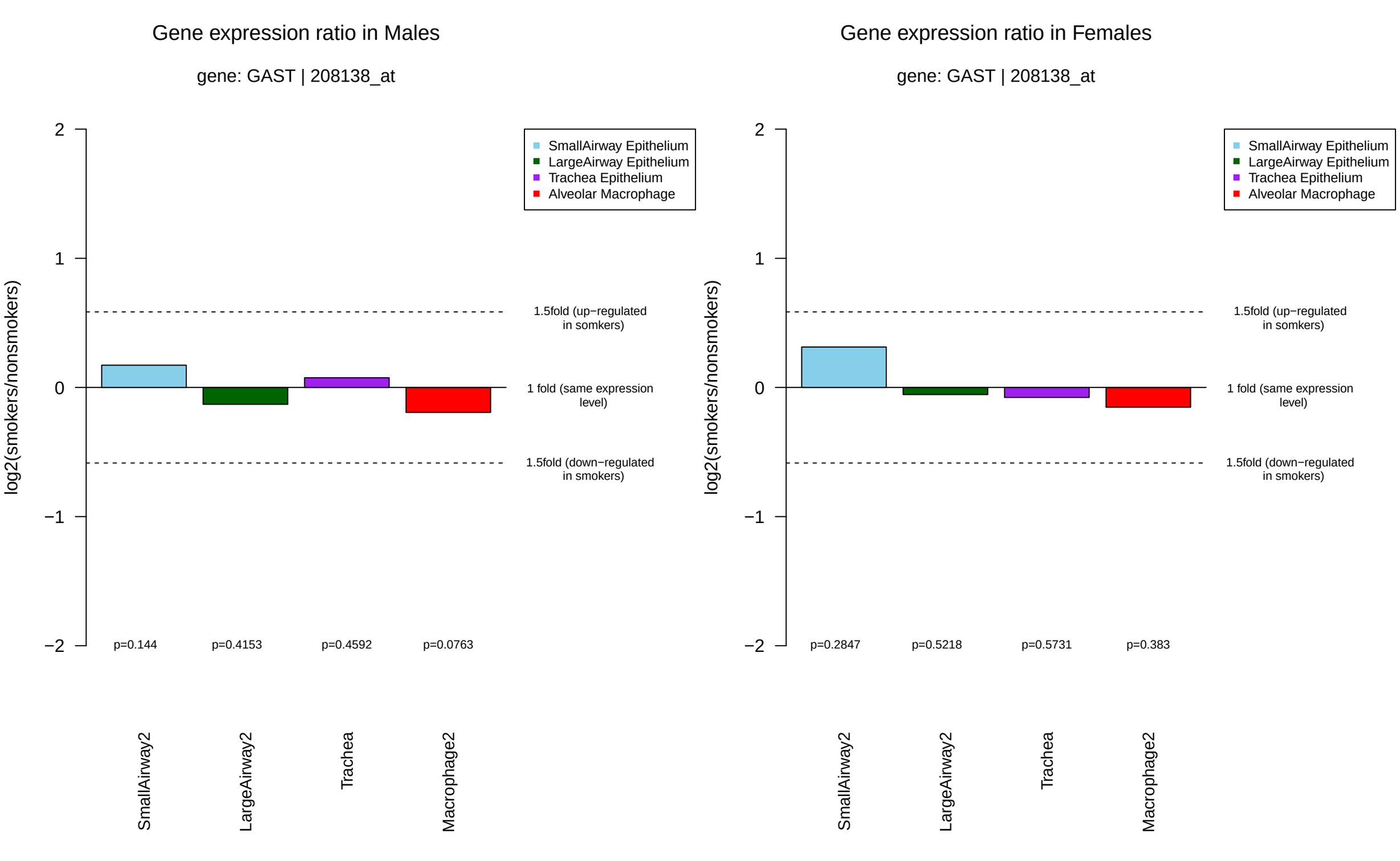
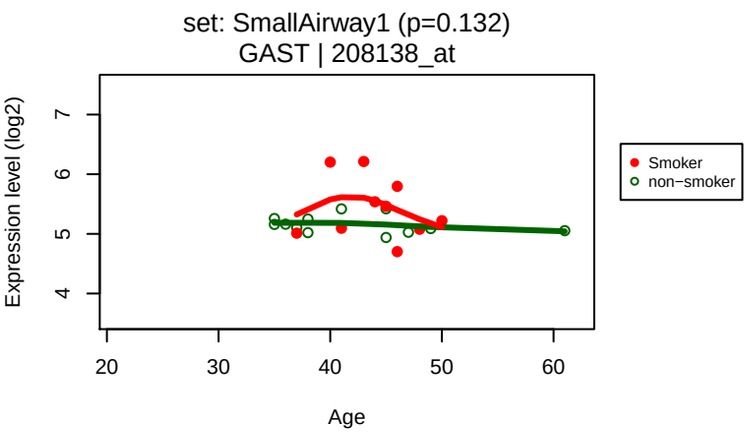
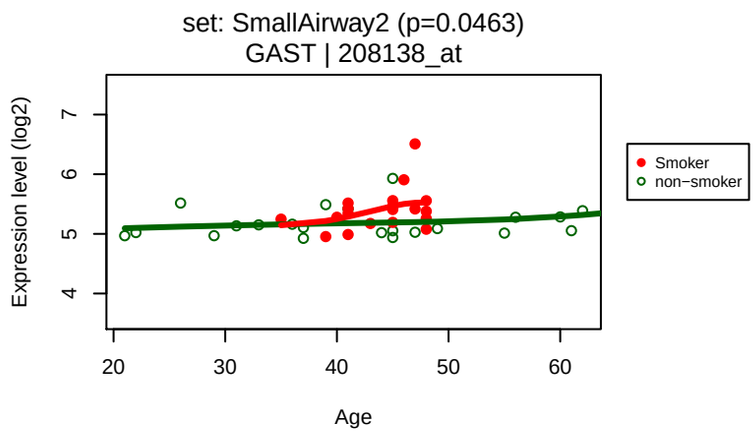
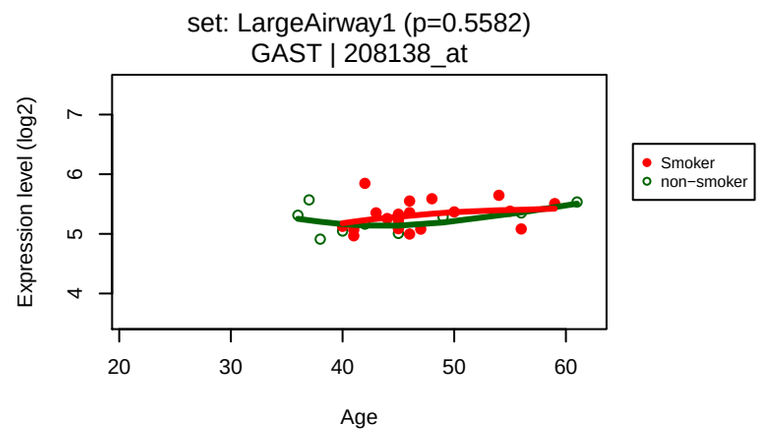

Gene expression of smokers and nonsmokers
gene: GAST | 208138_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.132
p=0.0463
p=0.0801
p=0.5582
p=0.4014
p=0.7482
p=0.9204
p=0.0539
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

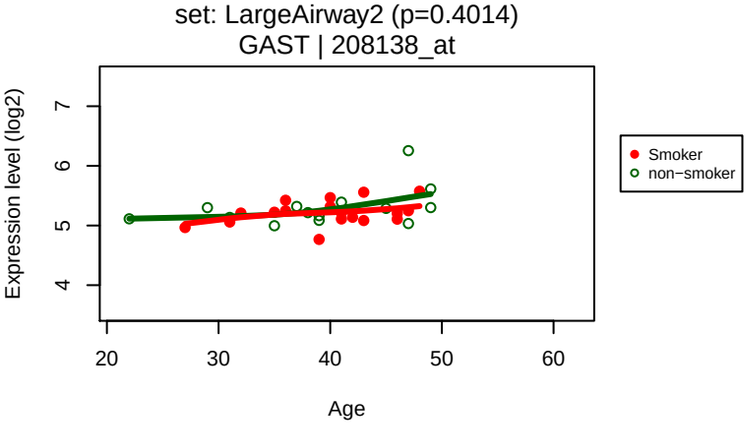

Gene expression ratio (smokers/nonsmokers)
gene: GAST | 208138_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.132
p=0.0463
p=0.0801
p=0.5582
p=0.4014
p=0.7482
p=0.9204
p=0.0539
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

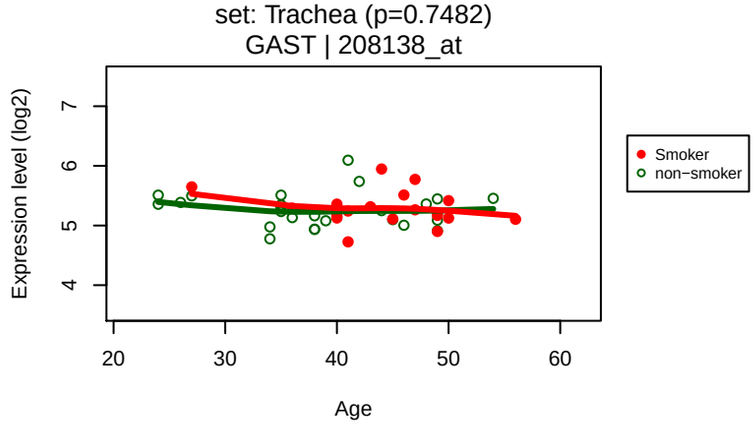

Gene expression ratio in Males
gene: GAST | 208138_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.144
p=0.4153
p=0.4592
p=0.0763
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GAST | 208138_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2847
p=0.5218
p=0.5731
p=0.383
SmallAirway2
LargeAirway2
Trachea
Macrophage2

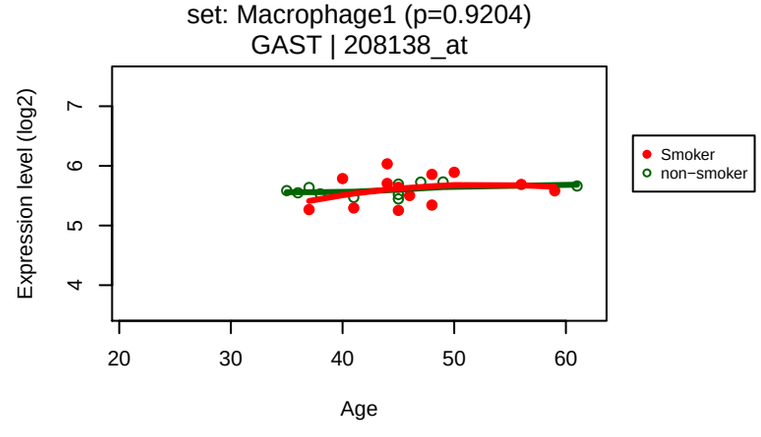

set: SmallAirway1 (p=0.132)
GAST | 208138_at
Expression level (log2)
Age
Smoker
non−smoker

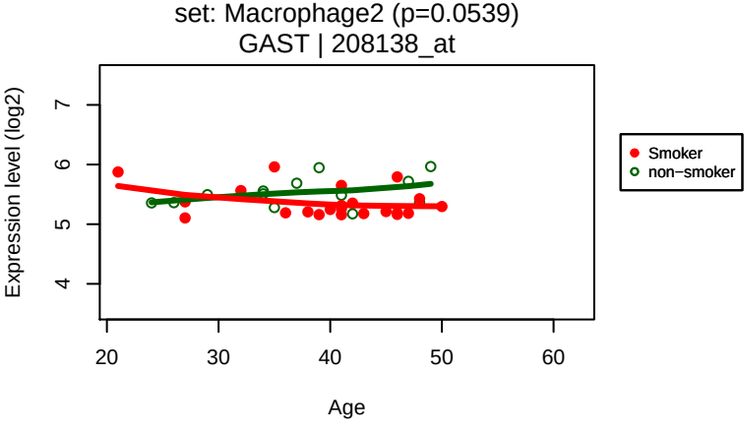

set: SmallAirway2 (p=0.0463)
GAST | 208138_at
Expression level (log2)
Age
Smoker
non−smoker

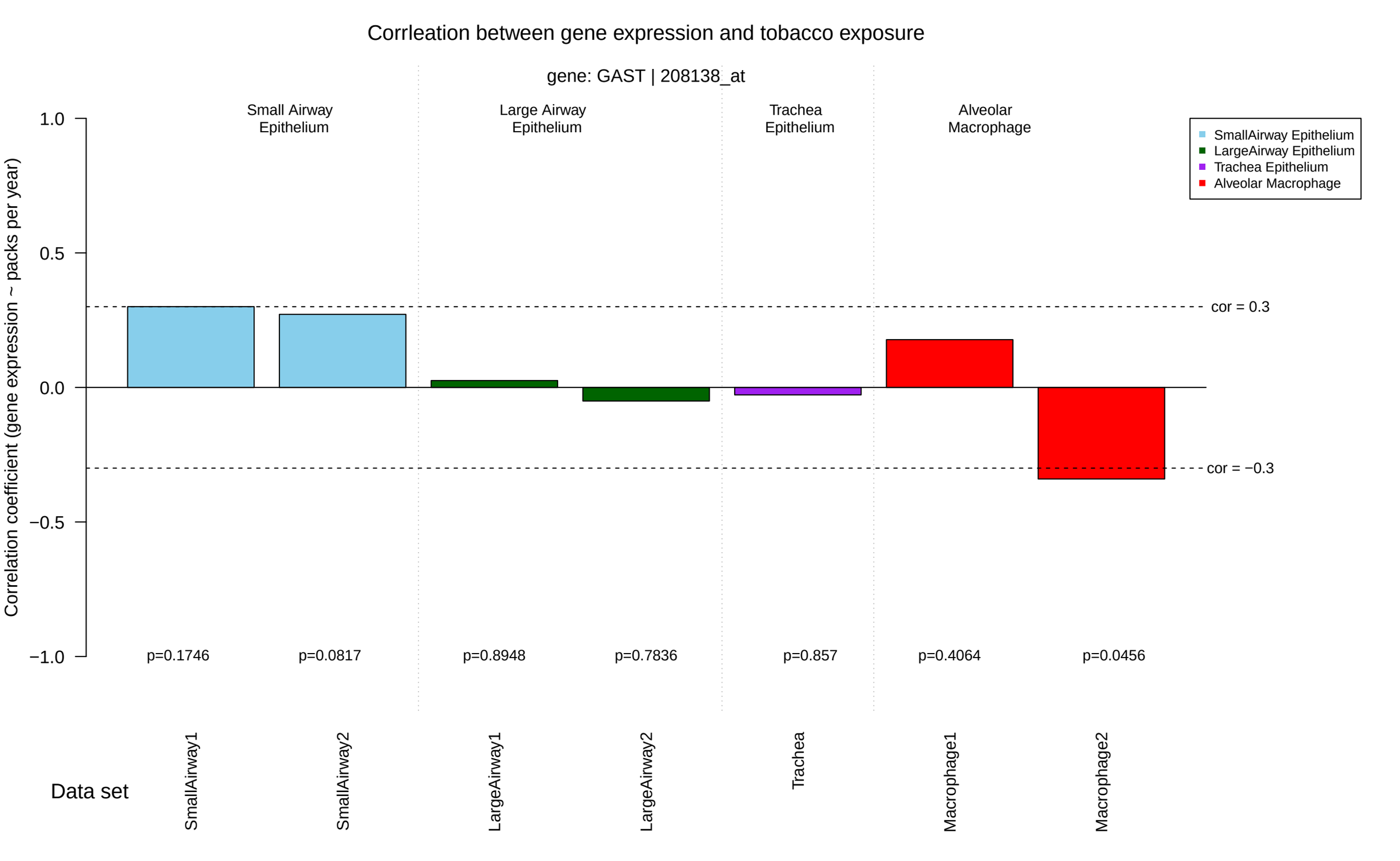

set: LargeAirway1 (p=0.5582)
GAST | 208138_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4014)
GAST | 208138_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7482)
GAST | 208138_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9204)
GAST | 208138_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0539)
GAST | 208138_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GAST | 208138_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1746
p=0.0817
p=0.8948
p=0.7836
p=0.857
p=0.4064
p=0.0456
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2